Purpose: CBSM entered into work agreement with IPC to review and provide feedback about a researchreport concerning three pilot studies fielded in different countries (Myanmar, Ecuador, and the Central African Republic (CAR)) to conduct a modal survey experiment to ascertain differences in outcomes of food insecurity measures between responses collected via face-to-face interviews and responses collected remotely via phone/cell interviews. CBSM was also asked to respond to commentary providedby Michelle Lacey.

Summary: Overall, the three pilot studies and their accompanying analyses showed merit in the face of limited resources to field a survey experiment such as this in a challenging environment with targeted survey participants who may be reticent to cooperate or difficult to contact in the first place. However, there was a general lack of information in this report required to properly assess the strength and validity of the three pilot studies and subsequent data analysis. For instance, there was scant information regarding the construction of the frame used for respondents in each country, and there waslittle detail provided about differences in survey administration about each treatment aside from the basic modal difference (face-to-face vs. phone) – details about by-mode differences in the survey instruments, interview protocols for field representatives, and cultural considerations taken to inform said protocols would be very helpful context for the reader. Further detailed commentary is provided below in three broad segments: sample/experimental design; study design; and statistical analysis.
Responses to Lacey's commentary round out this review document.

Feedback on Pilot Study Report:

Comments regarding sample/experimental design:

1. It appears that there is a difference in terminology used by the researchers regarding sample design for the three pilot studies in this report. In all three studies, a sample frame was stated tohave been established to conduct each experiment, but all households in the frame were then used in the experiment in some way – thus, no sample was actually drawn, and the sample design is, in effect, a census of the frame. There are valid questions raised by the expert reviewers that we would echo regarding the appropriateness of the ad-hoc frame construction relative to a representative sample of the population of interest, but as stated by the study administrators, that was not the aim of these pilot studies.

   Response: *The observation of the CBSM regarding the drawing of the sample of the study in the initial 3 countries partially correct. As stated in the report (see page 9), the 3 countries used different universe as sampling frames. This is to allow to test how the rCARI methodology performs amongst different context and with different populations. In* ***Myanmar****, the sampling frame was restricted to only IDPs who are all WFP beneficiaries. The main reason that informed the decision to use only IDPs in Myanmar was the lack of cooperation from local authorities and field access for household listing and subsequent face-to-face interviews. However, in Ecuador and CAR, the study sample was sampled through a random walk because there were not up to date census data available and other considerations like budget and security concerns. Thus, all listed households were same as sampled households.*

   In ***Ecuador****, all households in the 27 villages (Admin5 areas) that were randomly selected from the 3 Admin1s formed the selecting frame. The original sampling*

approach was to conduct a village census before drawing the sample however, due to budgetary, time and other security considerations, we adopted a random-walk approach to select and enlist households into the study. Starting from a known point, in each selected Admin5, and using an interval of 3, enumerators randomly selected households. During the listing phase, enumerators with informed consent of the household head collected some baseline information that included the phone number (at least 2 or more numbers within each household), physical address of the household (GPS location), name, gender and age of household head and time of availability for the interview. Households that refused to participate were skipped and the next household enlisted. Phone ownership in Ecuador at the time of the study was about 80% meaning most of the listed households had access to at least one working mobile phone. Households that were enlisted but had not access to mobile phones were excluded from the sample before randomization process (See page 9 and Figure 2 in comprehensive report for details). The methodology implemented in Ecuador was replicated in ***Central African Republic (CAR)*** *with a slight modification. The random walk method was followed to enlist the study households. However, based on lessons learnt from the Ecuador pilot, the listing was done using an expanded questionnaire that collected more baseline socioeconomic characteristic of the listed households. Furthermore, because of bad network coverage, we included a filter for identifying households that lived in areas without phone network. Secondly, we wanted to investigate the food security outcome of households that are missed in traditional phone surveys but giving such households phones to participated in the remote survey. Thus, households that did not have phones but lives in localities with mobile phone coverage were retained and used for the randomization process (see page 10 and figure 3 for the full details).*

The team will consider doing a full village level census for the upcoming pilot in Burkina Faso to address this concern raised by the US CSBM team. However, it is worth noting that this will have cost implication for data collection since more enumerators and time will be required to conduct a full village census.

2. If the intent regarding the request to assess the "sample design" is actually in part to be an assessment of the experimental designs for each of the pilot studies, that's within the realm of CBSM to garner an informed opinion. The simple randomized assignment of households into either the Remote (test treatment) or F2F (control treatment) in both the Myanmar and Ecuador pilots seems reasonable enough, though it is curious that there was no attempt to control assignment on key demographic or socioeconomic characteristics to ensure sufficient representation of sub-groups across treatments. The CAR pilot, with its multi-stage treatment assignment method, demonstrates a fine attempt to correct for low-SES participants in the Remote group.

   Response: *In Myanmar, because the sample frame was IDPs register from WFP operations in Kachin and Shan Admin1 Areas (no formal listing was done) and the fact that the team dd not receive local approvals to conduct household listing, no baseline household information such as demographic characteristics were available prior to the data collection phase. However, we perform a statistical balance test after the data collection was done and sample largely showed that the two study groups, i.e., F2F and remote were statistically similar based on observable household characteristics (such as age, gender, education).*

In Ecuador and CAR, observable household demographics were collected during the listing and used to perform a test of statistical comparability based on the observable household characteristics (such as age, gender, education, and household size) of the two study groups before field interviews started. Randomization was only stopped and considered stable when the two groups were considered statistically balances, i.e., no significant statistical differences observed for these variables between the two groups. However, given that the two groups will not remain constant even if statistical comparability is achieved during randomization because of attrition (some households did not participate after initially giving consent, others were not reachable and others did not complete the respective interviews), it is the comparability of households/respondents that completed the surveys in the two groups that should be consider. Therefore, we have conducted post attrition analysis of the comparability between the two groups (see Appendix Tables 25, 26, 27).
However, these concerns of CBSM have been considered and integrated into the impending pilot in Burkina Faso and the subsequent pilot studies. The randomization into the different study groups will be implemented using key demographic clusters to ensure that each different sub-group in the final sample is evenly distributed between the two study groups.

3. The other element of sample frame construction worth mentioning is the size of the frame. It is remarkable that the overall response rates from these three pilot studies is so high (60 percent or more). And yet, the size of the resulting response groups in each treatment are small enough such that apparent modal differences in the various food insecurity measures across two of the three pilot studies are not statistically significant (e.g., Table 3). This is most apparent in the Myanmar pilot where half of the Remote cases were lost, but it can also be observed in the Ecuador pilot. For Ecuador, the effect of doing a second round of sampling to shore up low response rates in the first round brings those results in line with Myanmar – it begs the question of whether an overall increase in the size of the sample frame would allow the experiment to pick up modal deltas as statistically significant rather than noise due to variability.

   Response: *Obtaining an informed consent from the listed households coupled with the community entry methods are the reasons for the high response rates observed in the study. Unlike traditional remote surveys that employs RDD (random digit dialing) we are using phone numbers that have been provided by households themselves with time slots indicating their availability. The issue of the sample size has been discussed extensively and agreed that in subsequent pilot countries (e.g., Burkina Faso) a larger sample size be considered to allow for meaningful analysis to also after attritions. In Burkina Faso for example, based on the CO previous data collection experience, double the number of households will be sampled.*

4. The CAR case is intriguing in that the higher disparity of food insecurity levels could partly be attributed to the unique experimental design that provided inexpensive cell phones to Remote- treatment households to enable their participation in the survey. One can reasonably assume that impoverished households that are more likely to experience food insecurity are also less likely to own cell phones with active service, so the administrators' efforts to bring cell phone access to those households have brought into the study data these households' influence in a way that is not reflected in the Myanmar or Ecuador pilots.

Response: *This observation is accurate. This is why a separate analysis of the CAR sample has been conducted taking into consideration these valid observations. This report is attached as the* ***CAR_rCARI_Report 2.2*** *for your perusal. In this report, as discussed earlier, the objective to be able to observe the food security outcome of households that would have otherwise been excluded in a traditional remote of CATI interviews. Therefore, the reports highlight the difference specifically comparing the phone owners with non-phone owners as well as with households that received phones separately. Furthermore, the report presents a number of practical challenges the team faced in the distribution of phones in the field (ranging from logistics, permits, all the way to respondents not reachable after receiving the phone) and the reason why such an exercise may not be successful in other pilots.*

5. In general, the experimental design involving selection of the phone respondents for CAR is described almost as an afterthought in the appendices, but it is in fact a key design decision that should be highlighted and explored with a comparative analysis into differences between the phone and non-phone groups.

   Response: *The selection of phone owners and giving phones to non-phone owners has been part of the original design from the beginning. The current report under review as mentioned is only preliminary hence the exclusion of these analysis in the main report. However, as explained above the full CAR analysis that includes these subgroups is now ready as* ***CAR_rCARI_Report 2.2.***

6. The lack of weighting to address sample skewness and low response rates were noted as a weakness of the study. If the authors could apply known information about the population the frame is based upon to craft rough weights, the sample could be balanced on key demographics (e.g., age) to control for some of the selection bias.

   Response: *The lack of reliable population level demographics and the fact that the current study is not to generalize the result to the universal population. This will be considered as the analysis progresses. Furthermore, in the subsequent pilots starting from Burkina Faso, there are plans to conduct full census of all Admin5 areas that the study will be implemented in during the listing phase. Baseline demographic characteristics of all households in the census will also be collected. A sub sample of the households will be included in the study and the weights constructed using the census data will be used on the sample.*

Comments regarding study design:

7. One thing the paper doesn't address is any design plans that were made to reduce any potential data collection challenges in either mode, including those that may be cultural. What information were people provided before the remote interview? Did you schedule times for the interviews where you knew the individual would be in locations with good network coverage? Were there any gender considerations made with interviewers either face-to-face or over the phone? Are there other important demographic differences that need to be considered for these countries/regions for effective interviewer-based data collection (age, class, tribes, etc.).

   Response: *A number of strategies were adopted by the study team in each of the pilot countries to reduce possible challenges during data collection and increase the rate of*

participation of sampled households in the study. In general, enumerators were selected from within the regions and areas where the study is being implemented. This increased the reception of enumerators particularly in the F2F mode. It also helped overcome some possible cultural sensitivities that could arise if total strangers to the setting are used. Also, households that were listed in the listing phase of the study were sensitized about the study and made aware that they may be contacted afterwards either through a F2F visit or a phone called and asked some questions regarding the study. For Myanmar, the study used WFP field officers (in both F2F and remote surveys) who are well known amongst the beneficiaries. In terms of time schedule for interviews, each household selected 3 different time slots that they deemed suitable either for the F2F visit or remote phone call. However, aside these times provided by the sampled households themselves, we developed calling protocols which were used by the enumerators based on empirical literature. On gender consideration of the enumerators, efforts were made to hire more female enumerators especially for the remote interviews since literature shows that the response rate tend to be higher for female enumerators than males in the context of remote surveys. For the F2F interviews particularly in CAR, enumerators were paired (females and males) as much as the numbers allowed. This further increased the acceptability during the community entrance process for the team.

8. There needs to be a more thorough discussion of the goals and limitations of the study, particularly regarding the universe being sampled within each country. The footnote on page 2 indicates one limitation within Myanmar. What about Ecuador and CAR? From what universe is the listing being conducted? Are these intended to be national samples? Or samples of smaller populations like WFP beneficiaries? I think it's fine to use special samples given budgets and other limitations, but the reader needs to have an idea of the external validity of the study to a broader population.

   Response: *In both countries (Ecuador and CAR), the sampling frame is restricted to universe of the Admin5 areas that meet the inclusion criteria and have been selected for the study. In Myanmar on other hand, the sampling universe was the IDPs list from 25 camps in Kachin and Shin regions (refer to page 9 &10 of the comprehensive report for details). The revised report has incorporated these concerns accordingly.*

9. Another important aspect of the design to mention would be anything that was done to reduce or eliminate costs for people who are completing via mobile phones. Explanation of how mobile phone costs work in these different countries would be helpful: Would it cost people who participated via the phone money to participate (e.g., how do mobile phone cost work in these countries - do people pay by the minute? Are the phone users only changed for outgoing calls and therefore completing the survey remotely is at no cost to the respondent?) If it would cost them money to participate, were they compensated for that?

   Response: *As it relates to cost associated with remote interviews, in all the pilot countries so far (including Burkina Faso), households are not charged any fee for receiving phone calls. Thus, participating in the study did not result in direct cost to the participating household. However, in line with CO operations in Ecuador, remote participants received a US$1.50 pay-out after completing the interview as a compensation for their time over the phone.*

10. Several questions about handling of phone numbers from potential respondents should be addressed:
    - How were the phone numbers provided? Were these gathered during listing?

      Response: *All phone number were collected during the listing phase except for Myanmar where the numbers were retrieved from the individual IDP camp managements.*

    - What if someone did not give a phone number? Were they automatically assigned to thecontrol group?

      Response: *As explained, for Myanmar and Ecuador, households that did not have at least one working phone number after the listing were dropped from the study (although these were minimal in both countries: 5 households in Myanmar and 8 households in Ecuador). In the Case of CAR, it was only households that did not have phone network coverage that were dropped while keeping households that did not have phone access but had phone network coverage (phones were provided to households that were assigned to receive remote interviews but did not own phones. See figure 3).*

    - Or were respondents only eligible for the experiment if they had provided a phonenumber? If so, that likely skewed the Face-to-face group based on phone ownership/network coverage.

      Response: *Yes, we only based the final sample for the experiment on households that provided at least one phone number for Myanmar and Ecuador. But in the case of CAR, households were eligible if they had phone network at their address. This was to cater for the phone distribution.*

Comments regarding statistical analysis:

11. It's not clear whether families of pairwise inference tests in these tabular results have been adjusted for simultaneous multiple comparisons. To use Table 3 as an example, we have in each pilot-study column a 4-way distribution of the food security measure by F2F, by Remote, and the delta between them. To properly assess the modal differences of each category, the researchers must adjust the individual confidence level of each comparison so that the overall confidence level of the family of tests is held firm at 90%, 95%, or whatever the researchers chose to base inference decisions (I don't believe this is ever stated). Also, another way to get at the question of "does mode affect response patterns on food security measures" as presented in this fashion would be a simple goodness-of-fit chi-square test at minimum, with a proper ANOVA exercise or generalized linear model with fixed and random effects as the preferred strategy. Neither seem to have been done in favor of descriptive analysis with basic pairwise inferential comparisons – not bad for initial stages of exploratory work, but more sophisticated methods should follow.

    Response: *On the issue of the statistical significance level of the study, this has been revised to reflect the following: * 95%, ** 99% and *** 99.9% significance levels*

respectively using a basic pairwise comparison. In terms of analysis of the data from the 3 pilot countries, this is still ongoing. For example, at this stage we have implemented a random regression model has been implemented using the CAR sample to isolate the effect of interview modality on food security outcome (defined as a binary variable that takes on the value of 1 if household is food insecure and zero otherwise). In this estimation a number of household level and enumerator characteristics were controlled for. The results from the estimated model are shown as Table 17 in the CAR Report. This will be replicated for Ecuador and Myanmar as well as for all the subsequent pilot countries data sets. The final project report will reflect the results from these regression analysis of all project countries.

12. It was noted that there was limited information about how the households in the study were determined to be "statistically identical" in order to make comparisons between, for instance, the CARI and rCARI indices.

    Response: *Based on the study design and the objective of comparing the outcome indicators between F2F and remote samples, we defined statistically identical to mean the composition of the two study groups, i.e., F2F and Remote were composed of households that are statistically similar based on respondent and household observable characteristics (age, gender of household head, marital status, household composition, income sources, etc.). See Annex Tables 25 & 26 for the statistical balance of Myanmar and Ecuador respectively.*

13. Regarding Table 1 (page 3): It would be more informative to break the columns of this table into to two components: face-to-face and remote. Then, report the response rate across the two conditions within the three countries and overall. Since there are response rate differences, this table should be followed with something like the tables in Annex I to test the equivalence of demographics across the conditions.

    Response: *The table has been revised accordingly and appears as Tables 1 and 2 for F2F and remote samples respectively. However, regarding the statistical comparison of the completed and not completed households, this will only be done for Ecuador and CAR pilots. The lack of baseline data for Myanmar will not allow such a comparison to be done. Going forward, this recommendation will be used as the template to report on the survey completion.*

14. It's not clear what "balanced households" means. Is this saying that the sample in CAR was only among phone owners regardless of the group? If CAR has one of the lowest mobile phone ownership, then people in CAR who have mobile phones are likely very different from the type of respondents who do not have mobile phones.

    Response: *Balanced households in this sense was referring to statistically comparable households. As noted, the sample composition is likely to be different given the economic status of phone and non-phone owners. This consideration has been explored separately in standalone report for CAR where the different households are compared depending on their phone ownership status. The results are reported in Annexes (Tables 19, 20, 21 and 22) of the CAR country report.*

15. In comparing the food insecurity measures by mode, it is not really clear to the reader what the differences in the questionnaires/protocol were between the phone and face-to-face group. Did you reduce the number of questions? Were the questions modified for mode? What was the average length of the interviews between phone versus face-to-face? It is difficult to interpret the results without having a good understanding of to what degree there were changes in the questionnaire due to mode.

    Response: *WFP uses standardized survey questionnaires to collect data for food security assessments. All of the questions required for CARI calculation were the same across the survey modalities except for expenditure module where the F2F questions has been modified or shortened to allow them to be administered over the phone. The F2F survey included additional modules that will be used for testing the correlation of the newly developed income source indicator against some other FS indicators that are not part of CARI or rCARI such as household hunger scale. Generally, the remote survey has few items compared to the remote. On average, it takes between 15-45 minutes and 45-90 minutes to complete a standard remote and F2F interview respectively.*

16. Why not compare CARI-Inc with rCARI-Inc in the analysis tables? All other comparisons were made. That might be a cleaner mode comparison if the measure of income source and change is not different across the CARI and rCARI. The FES comparison in Table 4 is somewhat confounded by the difference in the FES measures. Furthermore, the numbers in these tables should be in percentages to match the text. It's not clear why some numbers are bolded without asterisks.

    Response: *The comparison of the CARI-Inc and rCARI-Inc has been done (the results are represented in Table 8 in the rCARI Report 2.2).*

17. More detail would be helpful to help the reader understand the extent of differences. Perhaps the authors are asking about different items in CARI whereas in rCARI they are asking about all expenditures. Just listing the number of items and giving examples would be helpful context to know. Furthermore, it's important to know how the expenditure module was shortened. Were categories combined or were there certain expenditures not asked about in the shortened version?

    Response: *This has been explained in the revised comprehensive report (Page 26, under the module length and food security indicator reporting section of the report).*

18. Regarding Table 6: did the respondents actually overestimate their consumption, or are the estimates just higher on the phone than in-person? Could there be interviewer effects between asking this in-person versus on the phone? Could there be any social desirability in the responses?

    Response: *The wording should reflect the fact that respondents reported higher consumption rather than "overestimate" over the phone than in-person or F2F. On the possibility of interviewer effects, further analysis would be pursued to investigate such effects across all pilot countries. There is also the fact of social desirability which we intend to investigate more as the study progresses to other pilot countries.*

19. Regarding Table 7: the directionality of coping strategies across the three pilot studies is

interesting (in Myanmar and CAR, phone respondents were more likely to adopt more reducing coping strategies relative to F2F counterparts, but the opposite was true in Ecuador). It is possible there might be social desirability. It would be helpful to know if there are any cultural elements that could explain this difference.

Response: *As rightly pointed out, these observed difference may be driven by social desirability which couldn't capture quantitatively during the data collection. We have reached out to the Ecuador CO to ask for their expertise and firsthand knowledge of the study areas to explain these results (qualitatively).*

20. Regarding Table 8: It is unclear that the apparent differences in emergency coping strategies can be called "overreporting". It is certainly a large difference but there's no evidence cited that respondents are misreporting. Certainly, with something like illegal activities, it is highly likely that there is social desirability at play here. It may be easier for respondents to say they are participating in illegal activities to someone on the phone than someone who has shown up to their house for a F2F interview.

    Response: *The wording of the text has been since changed to reflect these concerns that may likely be driven by social desirability as is rightly pointed. In Burkina Faso, the working of this coping strategy will be re-considered. Particularly, more efforts will be put into the training of enumerators and field testing to reduce these noise as much as practicable.*

21. The intent and function of Table 11 (Impact of Survey Modality on Income) is unclear. Is it trying to show agreement between FES and income indicators? If so, we would suggest showing the percent agreement or some other measure like kappa across the countries.

    Response: *The income indicator is one of the variables that was collected using the same set of questions in both F2F interviews and over the phone. This table (now Table 13) shows how the module compares across the two modalities. All the numbers reported in here are in percentage of households that reported each income classification based on the new rCARI console.*

22. Due to the loss of half of the expenditure data in the Myanmar sample due to translation errors, we wonder if there were any potential context effects for questions administered after this module that we should be considering in the analysis. Was the FES module not translated at all? Or were there errors within some questions that render the composition of the index unusable?

    Response: *The error was in the non-food expenditure; it was translated to Myanmar language as food expenditure. So basically, the food expenditure was asked twice. The research team discovered this error during the data analysis of both food and non-food components of the expenditure module. However, the subsequent pilots in Ecuador and CAR had this error rectified because the error was only restricted to the translated version and not the generic module.*

<u>Response to Lacey Commentary</u>:

- Lacey described three key research questions that the studies ought to address when assessing the effects of the modal experiment upon the different food insecurity measures. While we agree these are important questions to investigate, we contend that the authors of the report did not adequately showcase efforts to address these research questions, instead focusing primarily on basic pairwise comparisons within each study as a largely descriptive analysis.
- We agree with Lacey that the identical construction of FCS, coping, and income measures between phone and face-to-face modes ought to result in easy, straightforward comparison of response profiles among these measures by mode, were it not for the listed confounding factors. These factors – geographic factors, livelihood variation, and enumerator effects – are all important considerations that appear to not have been accounted for in the analysis for the pilot studies.
- Lacey notes that the pilot studies were not intended to yield nationally representative results, and that the focus was strictly upon the modal effects upon response profiles relative to each of the food insecurity measures. We echo this statement, and we emphasize that the report did not make this important point explicitly. Furthermore, it is concerning that even without a focus on population-level estimation, there was no clear statement of the type of population that a full-scale survey of this type would aim to measure.
- Lacey correctly states that the simple pairwise comparisons in the study are a welcome starting point for analysis but should be followed with more comprehensive analysis such as mixed models with fixed and random effects. We concur with the assertion, as the types of overall statements made in the report cannot be supported with pairwise comparisons alone.
- The summary statement (below) is largely agreeable:

  > *In summary, the experimental designs proposed for data collection are appropriate for investigating the comparability of remote and face-to-face modalities, as well as for determining the impact of replacing the FES with the INC indicator. However, as is typically the case, the devil is in the details. Care must be taken to minimize confounding factors wherever possible, and where unavoidable (such as in the case of enumerators) to randomize as much as possible to enable analysis that can account for such factors. If this is done, then the results of these studies are likely to be both publishable and impactful.*

  However, we are not entirely convinced to Lacey's first sentence since the authors did not supply sufficient evidence to support this claim of appropriateness. It's possible that the work may have been done, but it's not stated explicitly within the report.

Overall response:

We are thankful to US CBSM for their comments and suggestions to improve and strengthen the rCARI validation study. As the study and data analysis is still ongoing, these recommendations will be critical in the next phase of the study. Most of the concerns raised in this documents relating to the sampling and study design have already been considered in revising the methodology for the pilot in Burkina Faso.

- In terms of sampling, the team will conduct a full census in each of the selected Admin5 areas using a standardized questionnaire which will collect more baseline characteristics of all households. The study sample will then be sampled randomly from each Admin5 census list. Also, the overall sample size will be increased by 50% to allow for more households to be interviewed. Also, the geographical locations and enumerator assignment will be improved to ensure confounding from these variables are minimized at least or eliminated at best.
- In terms of the implementation, aside randomizing at the Admin5 level as has been done previously, in the upcoming BF pilot this will be done using a number of household/household head level characteristics such as age, gender, education, and household size to ensure that the different categories are well represented in each of the study arms/groups.
- On the aspect of analysis, given that the report shared with BHA was a preliminary one, data analysis is still ongoing. A separate comprehensive report for CAR has been produced. The final report/s will include the suggestion given in this review.
- The full document showing the revised sampling methodology for the upcoming pilot in BF addressing the issues raised in this review is attached as BF_Sampling.